

The data mining process



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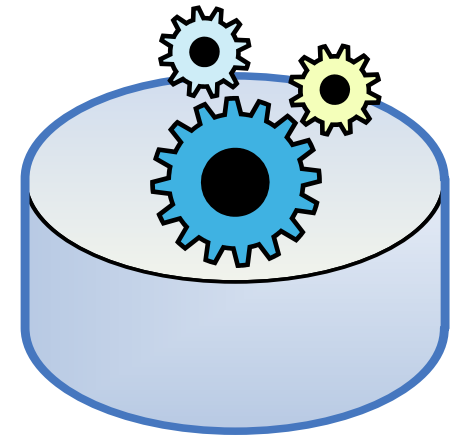
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Data mining →

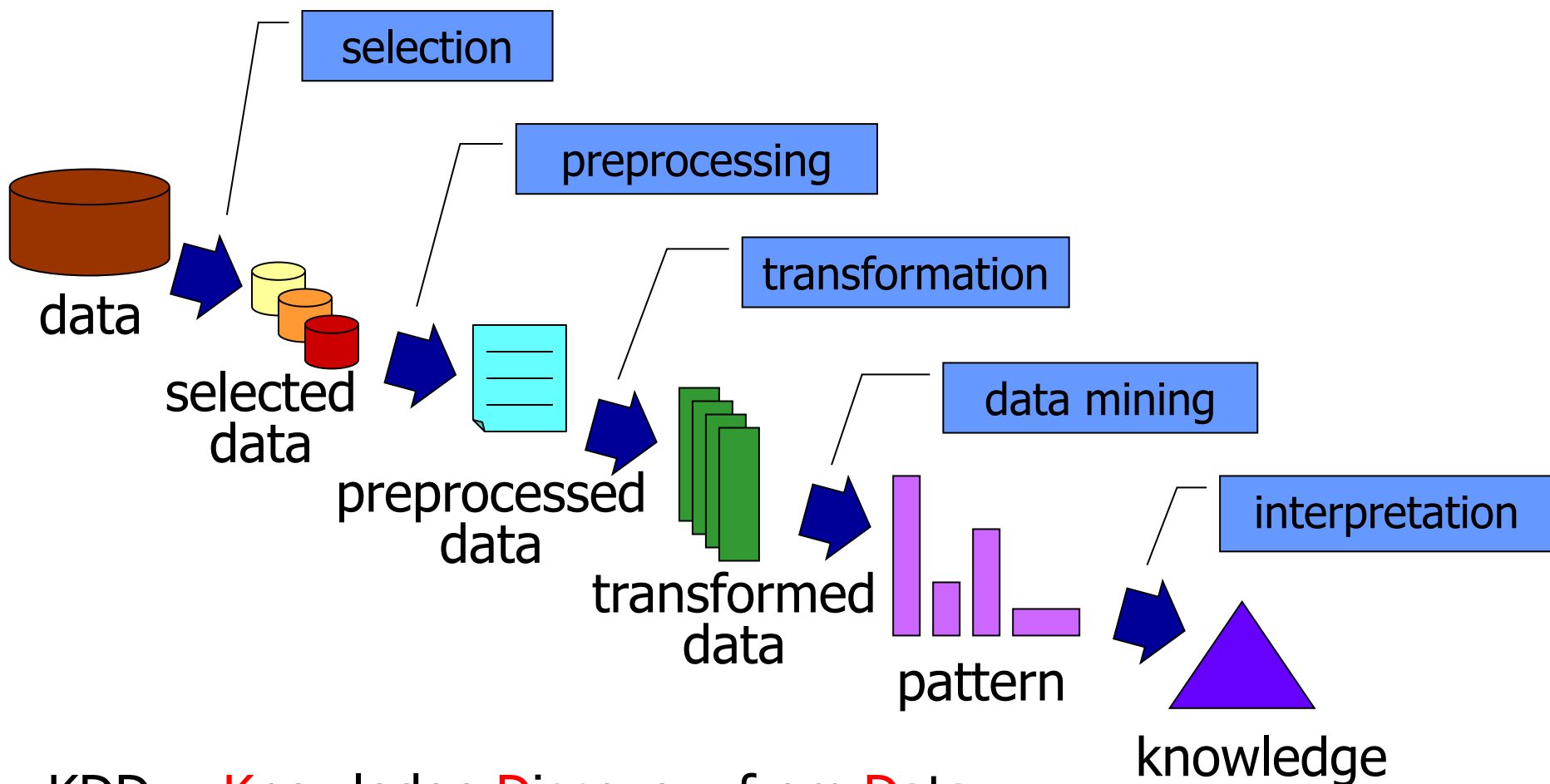
Algorithms that can be used to gain and explore information from data (machine learning)

- Non trivial extraction of
 - implicit
 - previously unknown
 - potentially usefulinformation from available data
- Extraction is automatic
 - performed by appropriate algorithms
- Extracted information is represented by means of abstract models
 - denoted as *pattern*





Knowledge Discovery Process



KDD = Knowledge Discovery from Data



Analysis techniques

- Descriptive methods → Classification
 - Extract interpretable models describing data
 - Example: client segmentation
- Predictive methods → Classification or regression
 - Exploit some known variables to predict unknown or future values of (other) variables
 - Example: "spam" email detection



Analysis techniques

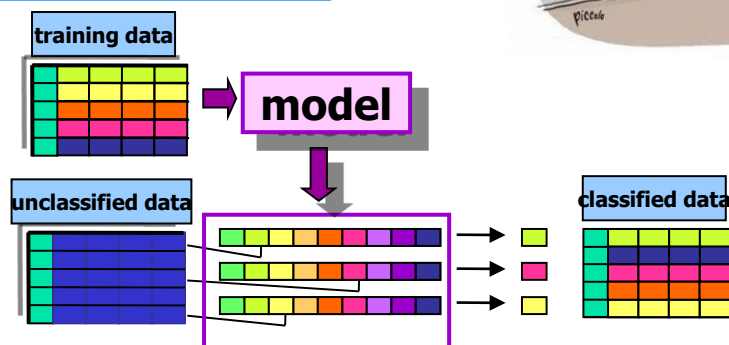
■ Association rule extraction

$X \rightarrow Y$
people who bought X also bought Y



■ Classification

The model compares properties of data already seen with the ones of the new one



■ Clustering

To organize objects in groups (clusters) based on the values of the features describing objects

